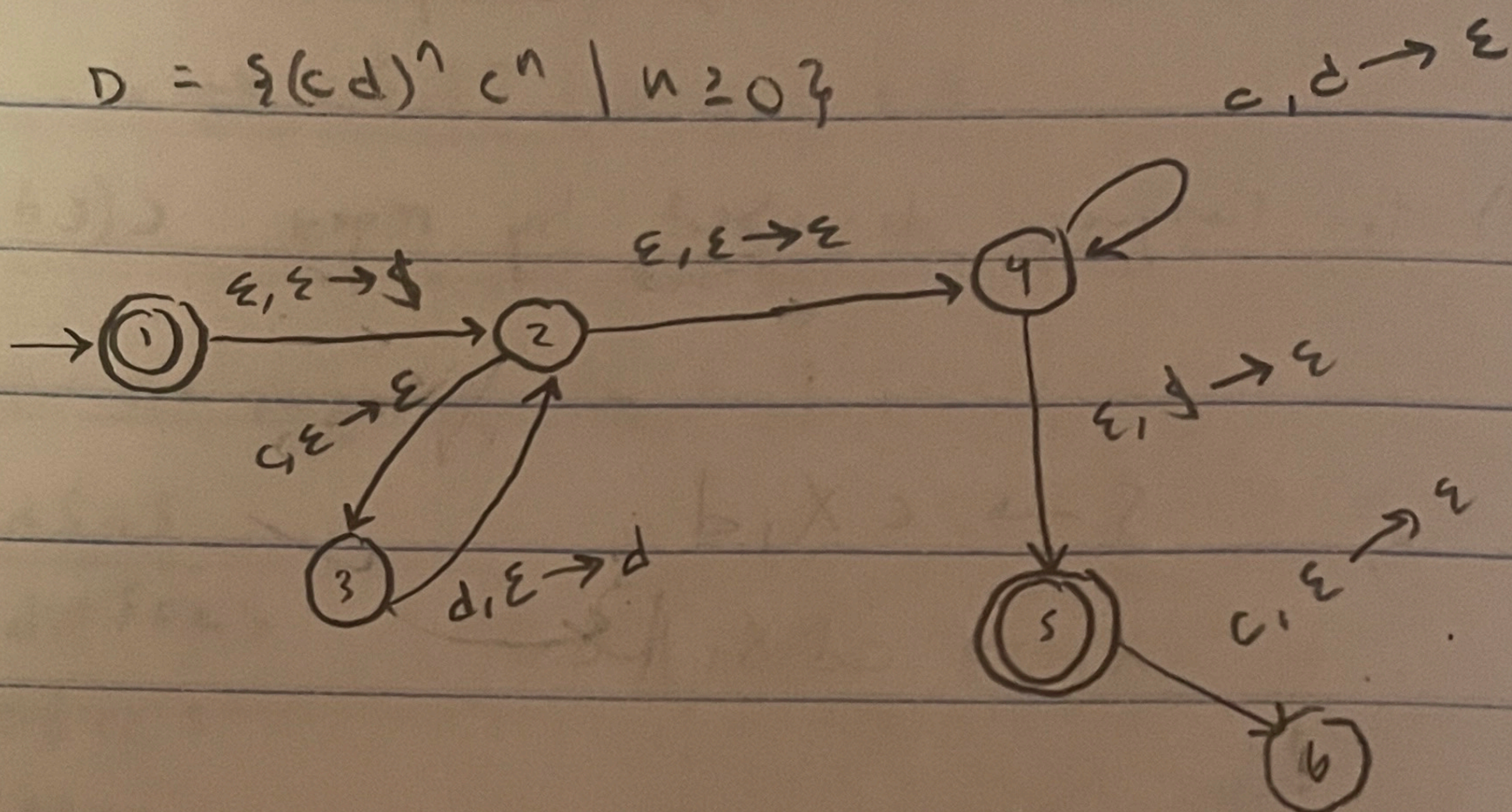


6

Write a pushdown Automaton for

$$D = \{ (cd)^n c^n \mid n \geq 0 \}$$

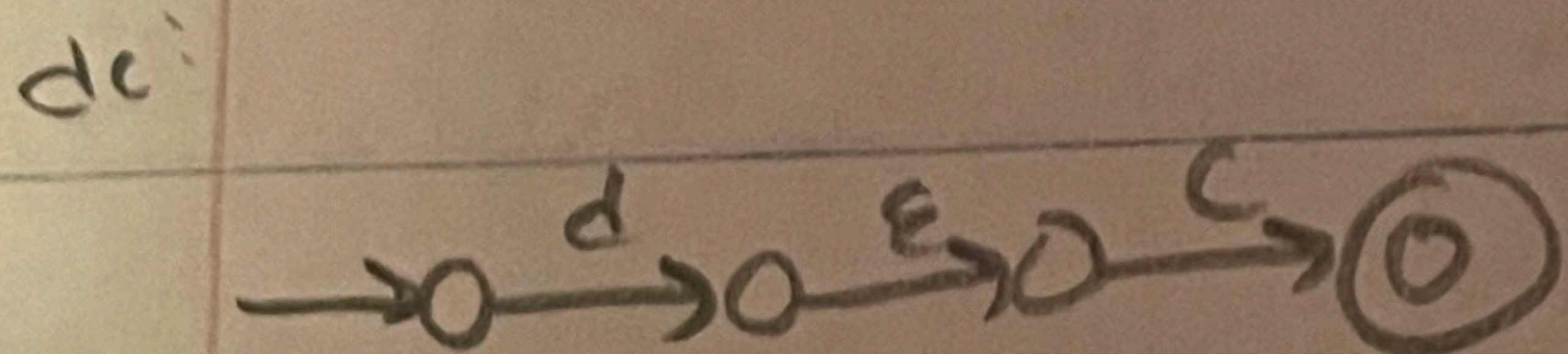
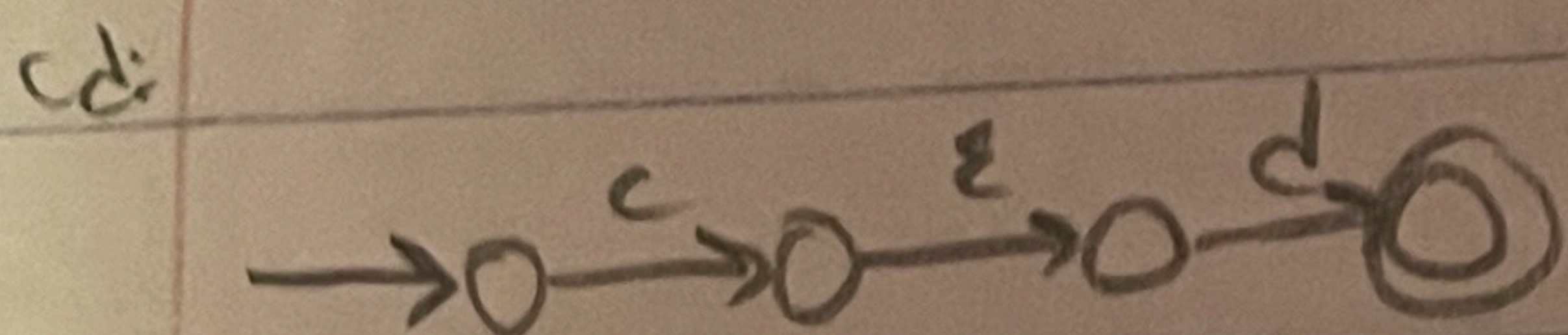
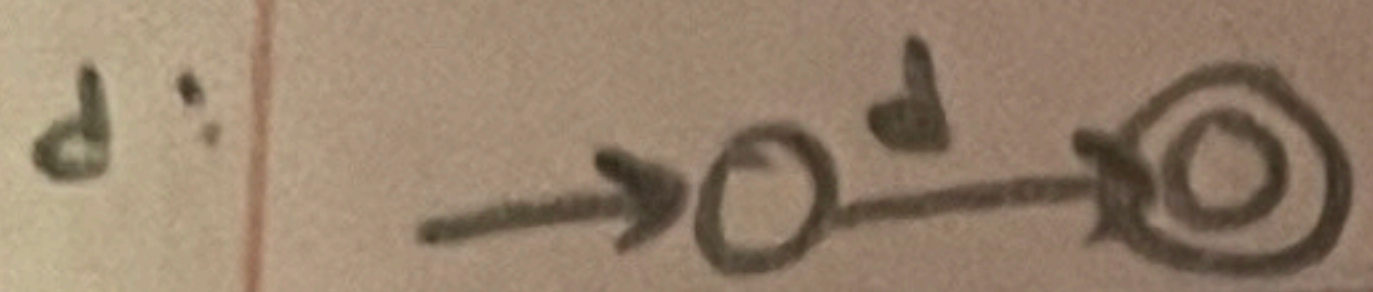
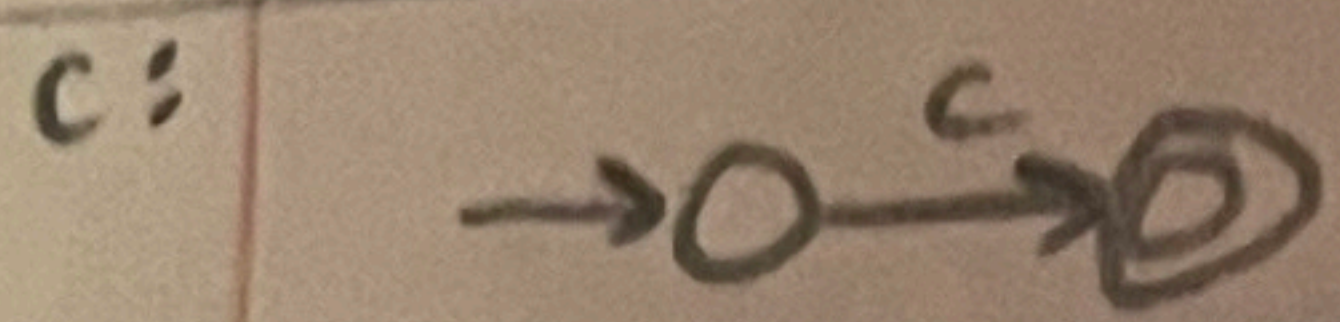


2/2

Added state 6 as
protection against strings

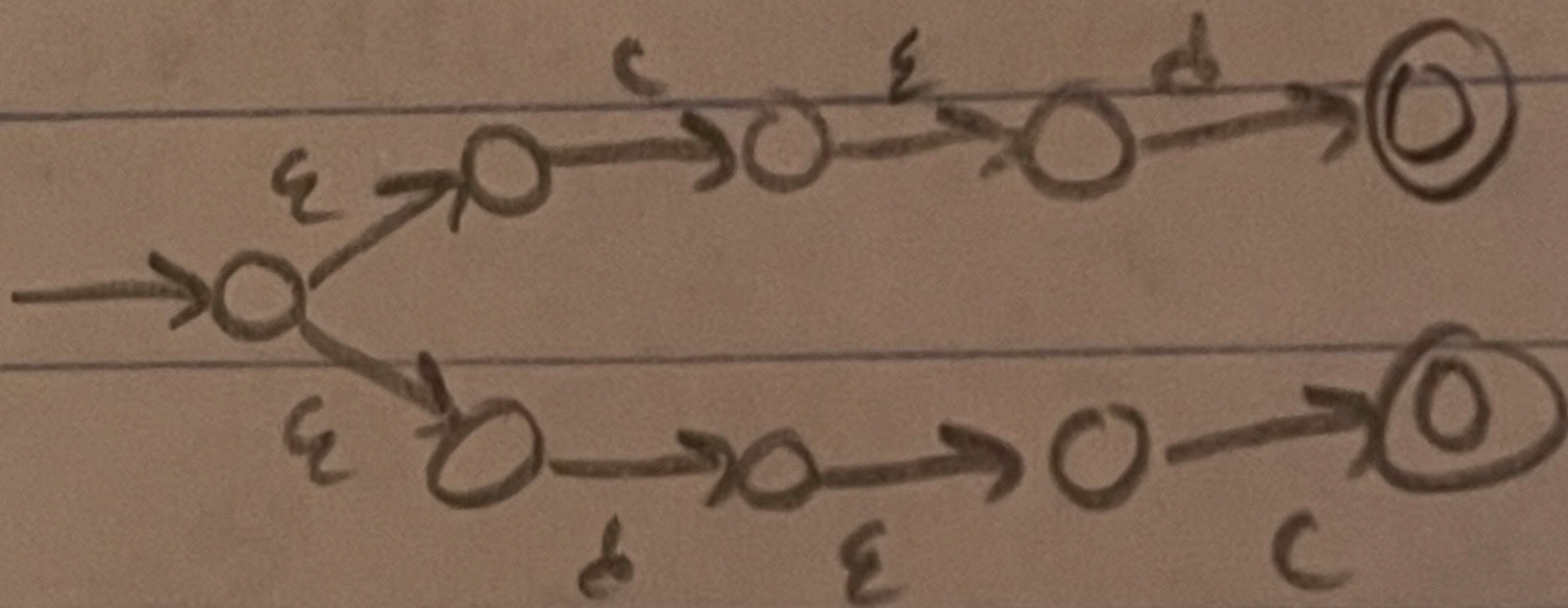
of form $(cd)^n c^{n+x}$, $x > 0$

2] Convert the regular expression $(cd \cup dc)^*$ into an NFA

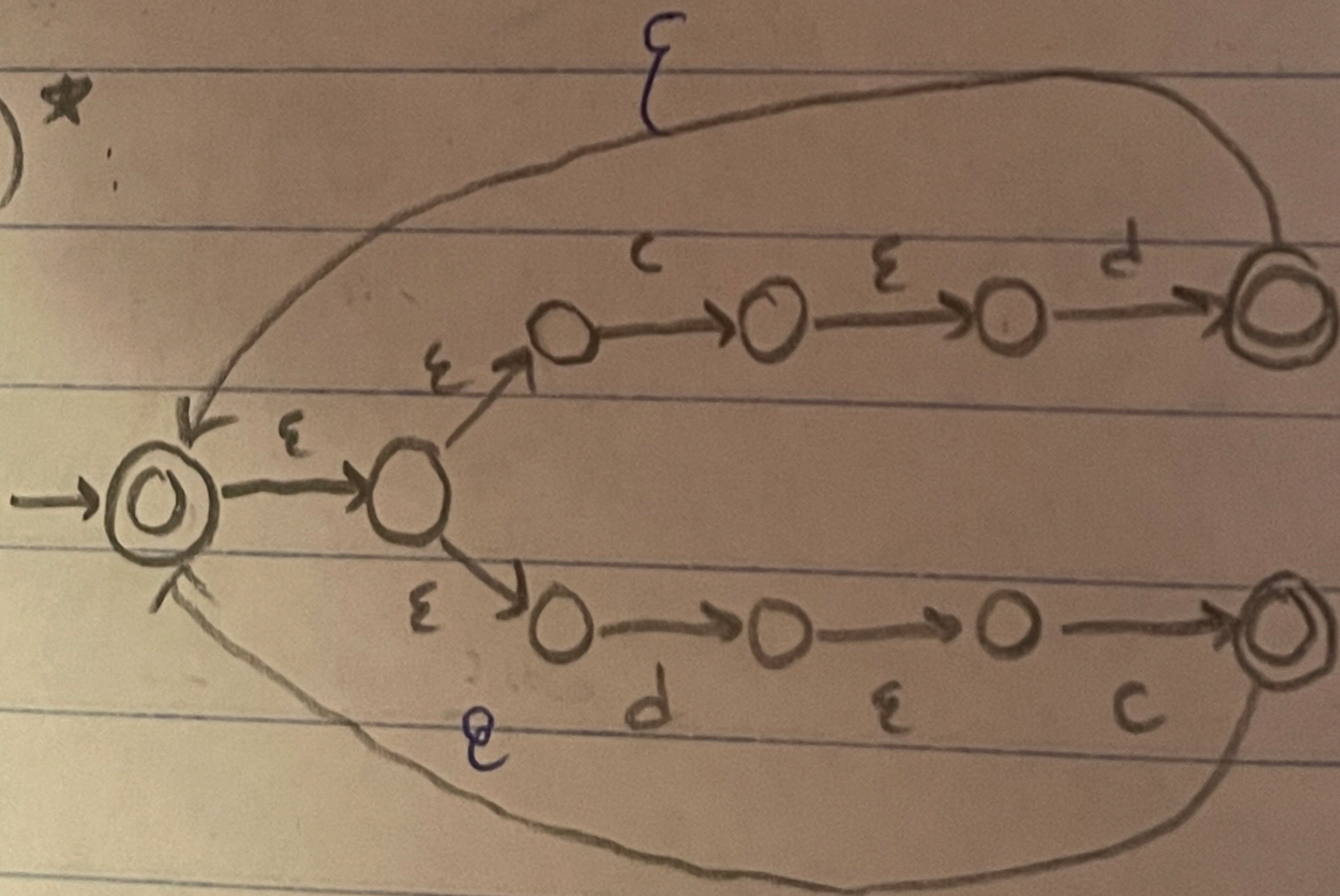


2/2

$cd \cup dc$:



$(cd \cup dc)^*$:



I have arrows going from accept of unioned term to start state generated by star operation. These could also point to original start of unioned terms, however I chose this to keep initials untouched.

4)

Consider the following CFG G :

$$S \rightarrow cSdS \mid dScS \mid \epsilon$$

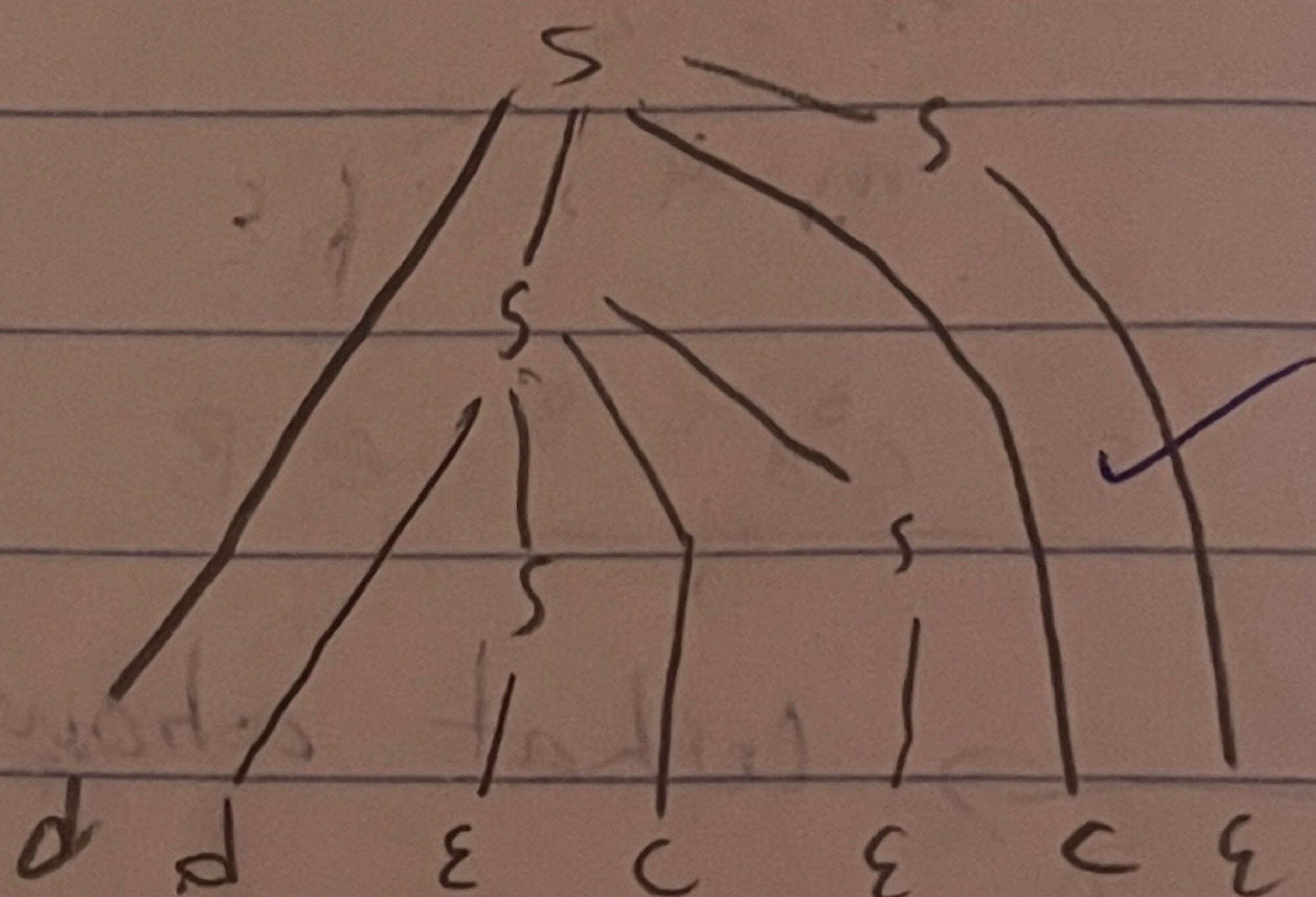
a) give the parse tree deriving $ddcc$

$$S \Rightarrow dScS$$

$$\Rightarrow ddScs$$

$$\Rightarrow ddccs$$

$$\Rightarrow ddcc$$



ddcc

yes.

* prof:
Part b does
not specify whether
or not parse tree
or derivation

b) show that the grammar G is ambiguous

string $cdcd$ is ambiguous

$$S \Rightarrow cSdS$$

$$\Rightarrow cds$$

$$\Rightarrow cdcds$$

$$\Rightarrow cdcds$$

$$\Rightarrow cdcd$$

derivation 2

$$S \Rightarrow cSdS$$

$$\Rightarrow cdscsds$$

$$\Rightarrow cdcds$$

$$\Rightarrow cdcds$$

$$\Rightarrow cdcd$$

These will yield different parse trees. using derivation to save time & space.

[5] For each of the following languages, give a CFG for it

a) the language described by regex $c(cd)^*d$

$$S \rightarrow cX_1d$$

$$X_1 \rightarrow cdX_1 \mid \epsilon$$

ex

cd

ccdd

ccddcd

b) $A = \{c^{3n}d^{2n} \mid n \geq 0\}$

ex ϵ $n=0$
 $cccd$ $n=1$
 $ccccccdd$ $n=2$

5/5

$$S \rightarrow cccSdd \mid \epsilon$$

c) $C = \{b^k c^{k+j} d^j \mid j, k \geq 0\}$

$$S \rightarrow X_{bc} X_{cd}$$

$$X_{bc} \rightarrow \epsilon \mid bX_{bc}c$$

$$X_{cd} \rightarrow \epsilon \mid cX_{cd}d$$

$j=k$ ϵ, bc, cd
 $j < k$ $bc, bbbcccd$
 $j > k$ $cd,$

10 min

10 min

$b^i c^{2i} d^i$

1 Give regular expressions that describe the following languages over alphabet $\Sigma = \{c, d\}$

a) $E = \{w \mid w \text{ has } ccd \text{ as a (possibly non-contiguous) subsequence}\}$

$$R_E = \Sigma^* c \Sigma^* c \Sigma^* d \Sigma^* -$$

b) $F = \{w \mid w \text{ has odd length, and its first and last symbols are different}\}$

$$R_F = c(\Sigma\Sigma)^* \Sigma d \vee d(\Sigma\Sigma)^* \Sigma c \quad \checkmark$$

2/3

③ Consider the language $B = \{c^3 d^m c^n \mid n \geq 2m \text{ and } m \geq 0\}$

a) give a string of length 15 that is in B

$$3 + m + 2m = 15$$

$$\leftarrow \text{let } n = 2m$$

$$3m = 12$$

$$m = 4$$

$$s = c^3 d^4 c^8 \in B$$

5/5

1/1

2.5/4

~~2.5/4~~ ~~3/4~~

→ What about other cases? y intersecting c^3

b) Show that B is not regular

or y containing a strict substring of d^4

Assume that B is regular. Therefore, there is P (pumping length) guaranteed by the pumping lemma applied to B.

Choosing string $s = c^3 d^P c^{2P}$ where $|s| > P$. Thus the following conditions must be satisfied:

1. $xy^i z$ for $i \geq 0$
2. $|y| > 0$
3. $|xy| \leq P$

Consider the case where y includes only 'd' symbols. An

example partition could look like this: $\underbrace{ccc}_x \underbrace{ddd}_y \underbrace{ccccccc}_z$. Knowing that

there must be at least twice as many 'c' symbols after 'd' symbols,

it is clear that repeating - or pumping - y containing only 'd' will generate a string with more 'd' symbols than 'c' symbols. Following

the contiguous 'd' substring, the pumped string will $\notin B$ as $n \geq 2m$ and $m \geq 0$. $\square$ Not regular

example $i=3 \rightarrow n=15$
 $|d|=9$
 $|c|=6$